

Aastha Agnihotri

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PROFESSIONAL SUMMARY

Fresher AI/ML Engineer specializing in Generative AI, RAG pipelines, and real-time ML systems. Built production-grade projects including a GNN-based Systemic Risk Platform with Kafka streaming and an Enterprise RAG Chatbot with citation-level accuracy. Seeking GenAI / ML Engineer roles in fintech or enterprise SaaS — available to join immediately.

SKILLS

AI & GenAI: RAG Pipelines, LangChain, Vector Embeddings, ChromaDB, FinBERT, GNN (PyTorch Geometric), Generative AI, Semantic Search

ML & Deep Learning: Machine Learning, Deep Learning, CNN, Neural Networks, Feature Engineering, MLflow

Programming: Python, SQL, Kotlin

Frameworks & Tools: FastAPI, Flask, React.js, OpenCV, Firebase, Groq API, Kafka, WebSocket

Data & Analytics: Advanced Excel, Power BI, BigQuery, ETL, Data Visualization

Cloud & DevOps: Cloud Deployment (Render), Real-Time Monitoring, Android Development

PROJECTS

Risk OS — Real-Time Systemic Risk Platform | Personal Project 2025

TechStack: GNN(PyTorchGeometric) • Kafka • WebSocket • FinBERT • MLflow • Gemini AI • FastAPI • RAG

- Designed a GNN-based contagion modeling engine to simulate financial network risk propagation across interconnected entities in real time — an approach not widely used in Indian fintech.
- Built real-time streaming pipeline using Kafka + WebSocket, enabling live risk score updates with sub-second latency.
- Integrated FinBERT for news sentiment analysis, feeding sentiment scores directly into risk models for dynamic adjustment.
- Implemented MLflow for experiment tracking and model versioning, following production-grade MLOps practices.
- Embedded a Gemini AI-powered analyst using RAG architecture to provide contextual risk explanations from financial documents.
- Deployed on Render with live dashboard — applicable to NBFCs, hedge funds, and SEBI-registered advisors.

Enterprise Knowledge Intelligence Assistant (RAG Platform) | Personal Project 2024

TechStack: React.js • FastAPI • LangChain • ChromaDB • GroqAPI • VectorEmbeddings • GenerativeAI

- Built a full-stack RAG platform transforming PDFs into queryable knowledge bases with citation-level accuracy (source + page number cited per answer) — reducing hallucination risk.
- Implemented semantic search using vector embeddings + ChromaDB for contextual Q&A; over financial reports, HR policy files, and resumes.
- Integrated LangChain + Groq API for LLM-powered natural language responses; users query documents in plain English.
- Developed React.js frontend with drag-and-drop PDF upload and FastAPI backend with real-time document ingestion pipeline.
- Directly applicable to enterprise document automation for CA firms, legal teams, and NBFCs.

DDoS Attack Detection & Protection System | Cloud-Based Project 2024

TechStack: Python • Render • AnomalyDetection

- Built system to detect and mitigate DDoS attacks using traffic pattern analysis and anomaly detection logic.
- Deployed on cloud (Render) for real-time monitoring; demonstrated cloud deployment and cybersecurity skills.

Medical Recommendation System | Academic Project 2023

TechStack: Python • RandomForest • Flask

- Built ML model using Random Forest for disease prediction with Flask-based web interface for real-time results.

Face Recognition & Sign-to-Text System | Academic Project 2023

Tech Stack: Python • CNN • OpenCV

- Implemented CNN + OpenCV based biometric authentication with gesture-to-text accessibility feature.

BhojanConnect—Food Ordering App | Academic Project 2023

Tech Stack: Kotlin • Firebase

- Developed Android app with real-time features and full backend integration using Kotlin and Firebase.

EDUCATION

B.Tech—Computer Science(AI&ML) | LokmanyaTilak College of Engineering, Mumbai 2022 – 2026

- Specialization: Artificial Intelligence and Machine Learning •
- HSC (2021) — Maharashtra State Board | 82.33%
- SSC (2019) — Maharashtra State Board | 74.20%

CERTIFICATIONS & TRAINING

- Python for Data Science — Google
- Data Science & Machine Learning Workshops — Acadbuddy
- Digital Marketing — IIDE

TARGET ROLES & AVAILABILITY

Open to: GenAI Engineer | ML Engineer | AI Engineer | Data Scientist (Entry Level)

Preferred sectors: Fintech • Enterprise SaaS • AI Startups

Availability: Immediate | Location: Mumbai (open to remote/hybrid)